Exercise – The Student's t distribution

1. Aim of the exercise

In this exercise we derive the stochastic representation and the PDF of the t statistic.

2. Stochastic representation of the Student's t statistic

In regression analysis, we are often interested in testing whether a population coefficient, such as β_1, is equal to some hypothesized value β_1^0. We formulate this as the null hypothesis and its alternative:

$$H_0 : \beta_1 = \beta_1^0$$
$$H_1 : \beta_1 \neq \beta_1^0.$$

We cannot test the null hypothesis directly because the population coefficient β_1 is unobserved. Instead, we use its sample estimate, $\hat{\beta}_1$, to draw inference about β_1. Specifically, we assess whether the observed estimate $\hat{\beta}_1$ differs sufficiently from the hypothesized value β_1^0 to reject the claim that $\beta_1 = \beta_1^0$.

$\hat{\beta}_1$ is a random variable and has a sampling distribution. This means that there is a probability associated with observing different values of $\hat{\beta}_1$ relative to the hypothesized value β_1^0. Consequently, the test of the null hypothesis must have a probabilistic nature. We therefore derive a test statistic that is itself a random variable and hence has a probability distribution.

The linear regression model in matrix form is given by

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}.$$

The vectors $\boldsymbol{y}$ and $\boldsymbol{\varepsilon}$ each contain n rows and one column. Let $\boldsymbol{X}$ be an $n \times K$ matrix, where each column contains the observations of one explanatory variable. Furthermore, let $\boldsymbol{\beta}$ be a $K \times 1$ column vector containing the coefficients $\beta_1, \beta_2, \ldots, \beta_K$.

To derive the t-statistic in a general setting, we consider an arbitrary coefficient β_k, where $k = 1, \ldots, K$. Its corresponding OLS estimate is denoted by $\hat{\beta}_k$, and the hypothesized value under the null hypothesis by β_k^0.

The OLS estimator is given by

$$\hat{\boldsymbol{\beta}} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y}.$$

Substituting the linear regression model into the OLS estimator yields

$$\begin{aligned}\hat{\boldsymbol{\beta}} &= (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'(\boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) \\ &= (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{X}\boldsymbol{\beta} + (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{\varepsilon} \\ &= \boldsymbol{\beta} + (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{\varepsilon}.\end{aligned}$$

If we assume

$$\boldsymbol{\varepsilon} \mid \boldsymbol{X} \sim N(\boldsymbol{0}, \sigma^2 \boldsymbol{I}_n),$$

then

$$\hat{\boldsymbol{\beta}} \mid \boldsymbol{X} \sim N\left[\boldsymbol{\beta}, \sigma^2 (\boldsymbol{X}'\boldsymbol{X})^{-1}\right].$$

That is, conditional on $\boldsymbol{X}$, the OLS estimator $\hat{\boldsymbol{\beta}}$ follows a Normal distribution with mean $\boldsymbol{\beta}$ and variance-covariance matrix $\sigma^2 (\boldsymbol{X}'\boldsymbol{X})^{-1}$. Since the test statistic is constructed from $\hat{\boldsymbol{\beta}}$, its

distribution is determined by the distribution of $\hat{\boldsymbol{\beta}}$. Because $\hat{\boldsymbol{\beta}}$ depends on the unobserved error term $\boldsymbol{\varepsilon}$, assumptions about the distribution of $\boldsymbol{\varepsilon}$ are needed. Here, we assume that $\boldsymbol{\varepsilon}$ follows a Normal distribution.

Subtracting $\boldsymbol{\beta}$ from both sides results in

$$\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} \mid \boldsymbol{X} \sim N\left[\mathbf{0}, \sigma^2 \left(\boldsymbol{X}'\boldsymbol{X}\right)^{-1}\right].$$

The t-test is a hypothesis test conducted for a particular coefficient estimator contained in the vector $\hat{\boldsymbol{\beta}}$. Let $\hat{\beta}_k$ denote the k-th entry of $\hat{\boldsymbol{\beta}}$. From the distribution of $\hat{\boldsymbol{\beta}}$, it follows that

$$\hat{\beta}_k - \beta_k \mid \boldsymbol{X} \sim N\left[0, \sigma^2 S^{kk}\right],$$

where

$$S^{kk} \equiv \left[\left(\boldsymbol{X}'\boldsymbol{X}\right)^{-1}\right]_{k,k}$$

denotes the k-th diagonal element of $\left(\boldsymbol{X}'\boldsymbol{X}\right)^{-1}$.

Under the null hypothesis, $\beta_k = \beta_k^0$. Substituting β_k^0 for β_k in the distribution of $\hat{\beta}_k - \beta_k$ yields

$$\hat{\beta}_k - \beta_k^0 \mid \boldsymbol{X} \sim N\left[0, \sigma^2 S^{kk}\right].$$

The fact that this distribution is centered at zero follows from the unbiasedness of the OLS estimator. Conditional on $\boldsymbol{X}$,

$$\mathbb{E}\left[\hat{\beta}_k \mid \boldsymbol{X}\right] = \beta_k.$$

Under the null hypothesis, $\beta_k = \beta_k^0$, and therefore

$$\mathbb{E}\left[\hat{\beta}_k \mid \boldsymbol{X}\right] = \beta_k^0.$$

Consequently,

$$\mathbb{E}\left[\hat{\beta}_k - \beta_k^0 \mid \boldsymbol{X}\right] = 0.$$

Therefore, under the null hypothesis, the difference $\hat{\beta}_k - \beta_k^0$ has mean zero and variance $\sigma^2 S^{kk}$.

The crucial step in the derivation is the transition to the distribution under the null hypothesis. The phrase "under the null hypothesis" means that we impose the condition $H_0 : \beta_k = \beta_k^0$. This condition is essential because it allows us to replace the unknown parameter β_k by the specific value β_k^0 stated in the null hypothesis. As a result, we obtain the probability distribution of $\hat{\beta}_k - \beta_k^0$ conditional on the null hypothesis being true. This distribution is called the "null distribution" and forms the foundation of the test statsitic. The construction of the test statistic therefore begins by deriving its probability distribution under the restriction imposed by the null hypothesis.

The next step is to standardize the distribution of $\hat{\beta}_k - \beta_k^0$. This is done by dividing by its standard deviation, $\sqrt{\sigma^2 S^{kk}}$. The purpose of this transformation is to obtain a distribution with mean zero and variance one, which allows us to compare the observed value to the standard Normal distribution. We therefore obtain

$$z_k \mid \boldsymbol{X} \equiv \frac{\hat{\beta}_k - \beta_k^0}{\sqrt{\sigma^2 S^{kk}}} \;\middle|\; \boldsymbol{X} \sim N(0,1).$$

The resulting distribution has mean zero and variance one. Because we have centered and scaled the distribution, it no longer depends on β_k^0 or σ^2. Conditional on $\boldsymbol{X}$, z_k therefore follows the standard Normal distribution.

One remaining problem is that the true variance σ^2 is not observed and therefore cannot be used directly. Consequently, we replace σ^2 with its unbiased estimator

$$\hat{\sigma}^2 = \frac{\hat{\boldsymbol{\varepsilon}}'\hat{\boldsymbol{\varepsilon}}}{n-K}.$$

This yields the statistic

$$t_k = \frac{\hat{\beta}_k - \beta_k^0}{\sqrt{\hat{\sigma}^2 S^{kk}}}.$$

where $\sqrt{\hat{\sigma}^2 S^{kk}}$ is the estimated standard error of $\hat{\beta}_k$.

Due to the replacement of σ^2 by $\hat{\sigma}^2$, the distribution of t_k is no longer immediately known. To determine its distribution, we first rewrite the statistic in an equivalent form. This rewriting will allow us to derive the distribution of the numerator and denominator separately. We obtain

$$\begin{aligned}
t_k &= \frac{\hat{\beta}_k - \beta_k^0}{\sqrt{\hat{\sigma}^2 S^{kk}}} \\
&= \frac{\hat{\beta}_k - \beta_k^0}{\sqrt{\hat{\sigma}^2}\sqrt{S^{kk}}} \frac{\sqrt{\sigma^2}}{\sqrt{\sigma^2}} \\
&= \frac{\left(\hat{\beta}_k - \beta_k^0\right)/\sqrt{\sigma^2 S^{kk}}}{\sqrt{\hat{\sigma}^2/\sigma^2}} \\
&= \frac{\left(\hat{\beta}_k - \beta_k^0\right)/\sqrt{\sigma^2 S^{kk}}}{\sqrt{\hat{\sigma}^2/\sigma^2}} \frac{1}{\sqrt{n-K}/\sqrt{n-K}} \\
&= \frac{\left(\hat{\beta}_k - \beta_k^0\right)/\sqrt{\sigma^2 S^{kk}}}{\sqrt{\left(\hat{\sigma}^2/\sigma^2\right)(n-K)/(n-K)}}.
\end{aligned}$$

We already know that, conditional on $\boldsymbol{X}$, the numerator of this fraction follows a standard Normal distribution:

$$\frac{\hat{\beta}_k - \beta_k^0}{\sqrt{\sigma^2 S^{kk}}} = z_k \sim N(0,1).$$

We now derive the distribution of the denominator,

$$\left(\hat{\sigma}^2/\sigma^2\right)(n-K).$$

First, we can write $\hat{\boldsymbol{\varepsilon}} = \mathbf{M}\boldsymbol{\varepsilon}$, where $\mathbf{M}$ is the symmetric and idempotent matrix defined as $\mathbf{M} = \mathbf{I_n} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$. Moreover, for two conformable matrices $\mathbf{A}$ and $\mathbf{B}$, we have that $\text{trace}(\mathbf{AB}) = \text{trace}(\mathbf{BA})$. This allows us to compute the trace of $\mathbf{M}$ as follows:

$$\begin{aligned}
\text{trace}\left(\mathbf{M}\right) &= \text{trace}\left(\mathbf{I_n} - \mathbf{X}\left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}'\right) \\
&= \text{trace}\left(\mathbf{I_n}\right) - \text{trace}\left(\mathbf{X}\left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}'\right) \\
&= n - \text{trace}\left(\left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}'\mathbf{X}\right) \\
&= n - \text{trace}\left(\mathbf{I_K}\right) \\
&= n - K
\end{aligned}$$

Second, we perform the following calculations:

$$\begin{aligned}\frac{\hat{\sigma}^2}{\sigma^2}(n-K) &= \frac{\hat{\boldsymbol{\varepsilon}}'\hat{\boldsymbol{\varepsilon}}}{n-K}\cdot\frac{1}{\sigma^2}(n-K)\\ &= \frac{\hat{\boldsymbol{\varepsilon}}'\hat{\boldsymbol{\varepsilon}}}{\sigma^2}\\ &= \frac{(\mathbf{M}\boldsymbol{\varepsilon})'(\mathbf{M}\boldsymbol{\varepsilon})}{\sigma^2}\\ &= \frac{\boldsymbol{\varepsilon}'\boldsymbol{M}'\boldsymbol{M}\boldsymbol{\varepsilon}}{\sigma^2}\\ &= \frac{\boldsymbol{\varepsilon}'\boldsymbol{M}\boldsymbol{\varepsilon}}{\sigma^2}\\ &= \left(\frac{\boldsymbol{\varepsilon}}{\sigma}\right)'\boldsymbol{M}\left(\frac{\boldsymbol{\varepsilon}}{\sigma}\right).\end{aligned}$$

From the normality assumption of the error terms, it follows that $\frac{\boldsymbol{\varepsilon}}{\sigma} \sim N(\mathbf{0}, \boldsymbol{I_n})$. If $\mathbf{x} \sim N(\mathbf{0}, \mathbf{I_n})$ and $\mathbf{A}$ is a symmetric and idempotent matrix, then $\mathbf{x}'\mathbf{A}\mathbf{x} \sim \chi^2(\text{trace}(\mathbf{A}))$. Since we know that $\mathbf{M}$ is symmetric and idempotent, we obtain

$$\left(\hat{\sigma}^2/\sigma^2\right)(n-K) \sim \chi^2(n-K).$$

The $n-K$ degrees of freedom have a natural interpretation. We started with n observations, but estimating K regression coefficients imposes K linear restrictions on the residual vector $\hat{\boldsymbol{\varepsilon}}$. This leaves only $n-K$ independent pieces of information in $\hat{\boldsymbol{\varepsilon}}$, which is why the Chi-squared distribution has $n-K$ degrees of freedom.

If $x \sim N(0,1)$ and $y \sim \chi^2(m)$ and x and y are independent, then the ratio $\frac{x}{\sqrt{y/m}}$ has the t distribution with m degrees of freedom.

We now show that

$$\frac{\hat{\beta}_k - \beta_k^0}{\sqrt{\sigma^2 S_{k,k}}}$$

and

$$\left(\frac{\hat{\sigma}^2}{\sigma^2}\right)(n-K)$$

are statistically independent.

Note that the terms

$$\frac{\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}}{\sigma} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\frac{\boldsymbol{\varepsilon}}{\sigma}$$

and

$$\frac{\hat{\sigma}^2}{\sigma^2}(n-K) = \left(\frac{\boldsymbol{\varepsilon}}{\sigma}\right)'\boldsymbol{M}\left(\frac{\boldsymbol{\varepsilon}}{\sigma}\right)$$

are statistically independent whenever

$$(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{M} = \mathbf{0}.$$

We therefore verify that this condition holds. Note that

$$\begin{aligned}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{M} &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\left(\mathbf{I}_n - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\right) \\ &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \\ &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \\ &= \mathbf{0}.\end{aligned}$$

This means that substituting the distributions of the numerator and denominator into the rewritten expression for t_k yields

$$t_k \sim \frac{N(0,1)}{\sqrt{\frac{\chi^2[n-K]}{(n-K)}}} = t\left[n-K\right].$$

This expression provides the stochastic representation of the Student's t-distribution with $n-K$ degrees of freedom. It shows how a t-distributed random variable can be constructed from a standard Normal random variable and an independent chi-squared random variable.

The preceding derivation establishes the finite-sample distribution of the t-statistic under the assumption of Normally distributed error terms. If we assume that

$$\boldsymbol{\varepsilon} \mid \mathbf{X} \sim N(0, \sigma^2\mathbf{I}_n),$$

then the t-statistic derived above has an exact finite-sample t-distribution. Consequently, the probability distribution of the t-statistic is exactly a t-distribution for every sample size n. If the error terms are not Normally distributed, the t-statistic no longer has an exact t-distribution in finite samples. However, under standard regularity conditions, the t-statistic converges to a standard Normal distribution as the sample size increases. Therefore, for sufficiently large samples, the finite-sample t-distribution provides a useful approximation for hypothesis testing.

3. PDF of the Student's t statistic

Let $f_T(\cdot)$ denote the function value, t the function input, and let ν denote the degrees of freedom. Then, the definition of the PDF of the $t(\nu)$ distribution is given by:

$$f_T(t) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)\cdot\left(1+\frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}}}{\Gamma\left(\frac{\nu}{2}\right)\cdot\sqrt{\nu\pi}}, \quad t\in(-\infty,\infty).$$

In this subsection, we show the proof for the probability distribution function of the $t(\nu)$ distribution using the approach of Hogg et al. (2019). Let $X \sim N(0,1)$ be a normal random variable and let $Y \sim \chi^2(\nu)$ be a Chi-square random variable, where X and Y are independent. Recall that we have the following probability density functions for random variables X and Y:

$$f_X(x) = \frac{1}{\sqrt{2\pi}}\cdot e^{\frac{-x^2}{2}}, \quad x\in(-\infty,\infty),$$

and

$$f_Y(y) = \frac{1}{\Gamma\left(\frac{\nu}{2}\right)\cdot 2^{\frac{\nu}{2}}}\cdot y^{\frac{\nu}{2}-1}\cdot e^{-\frac{y}{2}}, \quad y\in(0,\infty).$$

Due to independence, the joint probability density function of X and Y is given by:

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{\sqrt{2\pi}} \cdot e^{\frac{-x^2}{2}} \cdot \frac{1}{\Gamma\left(\frac{\nu}{2}\right) \cdot 2^{\frac{\nu}{2}}} \cdot y^{\frac{\nu}{2}-1} \cdot e^{-\frac{y}{2}} & \text{, if } x \in (-\infty, \infty) \text{ and } y \in (0, \infty) \\ 0 & \text{, otherwise.} \end{cases}$$

By definition of the t distribution, we have that a random variable T follows the t distribution with ν degrees of freedom if it has the following expression:

$$T = \frac{X}{\sqrt{Y/\nu}} \sim t\left(\nu\right).$$

To derive the probability density function of T, we will use the transformation technique and define $U = Y$. From this it follows that the equations

$$t = \frac{x}{\sqrt{y/\nu}} \quad \text{and} \quad u = y$$

define a transformation that maps $S_1 = \{(x,y) : -\infty < x < \infty, 0 < y < \infty\}$ one-to-one and onto $S_2 = \{(t,u) : -\infty < t < \infty, 0 < u < \infty\}$. We want to derive the joint probability density function $f_{T,U}(\cdot)$, which requires us to compute the appropriate inverse functions and the absolute value of the Jacobian of the transformation. It follows that the inverse functions are given by:

$$x = t\sqrt{\frac{u}{\nu}},$$

$$y = u.$$

This allows us to compute the Jacobian matrix by taking the first-order partial derivatives of the inverse functions with respect to t and u.

$$J = \begin{bmatrix} \frac{\partial x}{\partial t} & \frac{\partial x}{\partial u} \\ \frac{\partial y}{\partial t} & \frac{\partial y}{\partial u} \end{bmatrix} = \begin{bmatrix} \sqrt{\frac{u}{\nu}} & \frac{t}{2\sqrt{u\nu}} \\ 0 & 1 \end{bmatrix}$$

From this it immediately follows that its determinant, i.e., the Jacobian is equal to

$$|J| = \sqrt{\frac{u}{\nu}}.$$

To derive the joint density function of T and U, we will use the formula below.

$$f_{T,U}(t,u) = f_{X,Y}\left(t\sqrt{\frac{u}{\nu}}, u\right) \cdot |J|$$

From this we can write the joint PDF as:

$$f_{T,U}(t,u) = \begin{cases} \frac{1}{\sqrt{2\pi\nu} \cdot \Gamma\left(\frac{\nu}{2}\right) \cdot 2^{\frac{\nu}{2}}} \cdot u^{\frac{\nu+1}{2}-1} \cdot e^{-\frac{u}{2}\left(1+\frac{t^2}{\nu}\right)} & \text{, if } t \in (-\infty, \infty) \text{ and } u \in (0, \infty) \\ 0 & \text{, otherwise.} \end{cases}$$

The marginal density of T is then

$$\begin{aligned} f_T(t) &= \int_0^\infty f_{T,U}(t,u)\mathrm{d}u \\ &= \int_0^\infty \frac{1}{\sqrt{2\pi\nu}\cdot\Gamma\left(\frac{\nu}{2}\right)\cdot 2^{\frac{\nu}{2}}}\cdot u^{\frac{\nu+1}{2}-1}\cdot e^{-\frac{u}{2}\left(\frac{t^2}{\nu}+1\right)}\mathrm{d}u. \end{aligned}$$

To solve this, let us define $z = \frac{u\left(\frac{t^2}{\nu}+1\right)}{2}$ and note that $\frac{\mathrm{d}z}{\mathrm{d}u} = \frac{\frac{t^2}{\nu}+1}{2}$, which implies that $\mathrm{d}u = \frac{2}{\frac{t^2}{\nu}+1}\mathrm{d}z$. Performing the change of variables in the preceding integrand yields:

$$\begin{aligned} f_T(t) &= \int_0^\infty \frac{1}{\sqrt{2\pi\nu}\cdot\Gamma\left(\frac{\nu}{2}\right)\cdot 2^{\frac{\nu}{2}}}\cdot\left(\frac{2z}{\frac{t^2}{\nu}+1}\right)^{\frac{\nu+1}{2}-1}\cdot e^{-z}\cdot\frac{2}{\frac{t^2}{\nu}+1}\mathrm{d}z \\ &= \int_0^\infty \frac{2^{-\frac{1}{2}-\frac{\nu}{2}+\frac{\nu+1}{2}-1+1}}{\sqrt{\pi\nu}\cdot\Gamma\left(\frac{\nu}{2}\right)}\cdot z^{\frac{\nu+1}{2}-1}\left(\frac{1}{\frac{t^2}{\nu}+1}\right)^{\frac{\nu+1}{2}}\cdot e^{-z}\mathrm{d}z \\ &= \frac{\left(1+\frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}}}{\Gamma\left(\frac{\nu}{2}\right)\cdot\sqrt{\pi\nu}}\cdot\int_0^\infty z^{\frac{\nu+1}{2}-1}\cdot e^{-z}\mathrm{d}z. \end{aligned}$$

Recall the definition of the $\Gamma(\cdot)$ function, which is given by:

$$\Gamma(x) = \int_0^\infty z^{x-1}\cdot e^{-z}\mathrm{d}z.$$

It follows that the marginal density of T simplifies to:

$$f_T(t) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)\cdot\left(1+\frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}}}{\Gamma\left(\frac{\nu}{2}\right)\cdot\sqrt{\nu\pi}}, \quad t\in(-\infty,\infty).$$

4. Final notes

This file is prepared and copyrighted by Akash Boelens, Renata Istrătescu, and Tunga Kantarcı. This file and the accompanying MATLAB files are available on GitHub and can be accessed via this link. This exercise draws on one reference: Hogg, R. V., McKean, J. W., Craig, A. T., 2019. Introduction to mathematical statistics. Pearson Education, Boston, eighth edition. Available at: link.